

PROOF OF THE FOUR-STEP D-FINITE RECURRENCE FOR OEIS A278992

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ABSTRACT. Krasko and Omelchenko gave an explicit exponential generating function for simple chord-labelled chord diagrams. The OEIS entry A278992 records a four-step polynomial recurrence for its coefficients as a conjecture. We prove that recurrence. The proof embeds the two terms of the generating function in a three-dimensional rational differential system, obtains a third-order polynomial annihilator, and applies an exact differential-operator left multiple whose coefficient equation is the conjectured recurrence. We also resolve the starting-index issue: with the canonical value $a_0 = 0$, the recurrence holds for every $n \geq 4$ and requires $(a_0, a_1, a_2, a_3) = (0, 0, 1, 1)$.

1. INTRODUCTION

Let a_n count simple chord-labelled chord diagrams with n chords. Krasko and Omelchenko proved that the exponential generating function is [1, Theorem 4.3]

$$(1) \quad F(t) = \frac{1 + \sqrt{1 - 2t}}{\sqrt{1 - 2t}} e^{-1-t+\sqrt{1-2t}} - (2-t)e^{-t} = \sum_{n \geq 0} a_n \frac{t^n}{n!}.$$

The OEIS entry A278992 gives the same generating function and currently lists the recurrence below as conjectural [2]. Our result is as follows.

Theorem 1.1. *Let F be the formal power series in (1), where the square root is the unique one with constant term 1, and set $a_n = n![t^n]F(t)$. Then, for every integer $n \geq 4$,*

$$(2) \quad \begin{aligned} 0 = & (2-n)a_n + (2n^2 - 8n + 7)a_{n-1} + (6n^2 - 18n + 11)a_{n-2} \\ & + (n-1)(6n-11)a_{n-3} + 2(n-1)(n-2)a_{n-4}. \end{aligned}$$

The necessary initial values are

$$(a_0, a_1, a_2, a_3) = (0, 0, 1, 1).$$

Equivalently, in the native OEIS offset 1, one may start (2) at $n = 5$ with $(a_1, a_2, a_3, a_4) = (0, 1, 1, 21)$.

All manipulations below take place in formal power series at $t = 0$. Thus rational functions whose denominators have nonzero constant term cause no analytic or branch ambiguity.

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2. DIFFERENTIAL ANNIHILATORS

Put $q = 1 - 2t$, let $s = q^{1/2}$ with $s(0) = 1$, and define

$$E = e^{-1-t+s}, \quad U = E, \quad V = E/s, \quad G = (2-t)e^{-t}.$$

Then $F = U + V - G$. Differentiating $s^2 = q$ gives $s' = -1/s$, and hence

$$(3) \quad \begin{pmatrix} U \\ V \\ G \end{pmatrix}' = A \begin{pmatrix} U \\ V \\ G \end{pmatrix}, \quad A = \begin{pmatrix} -1 & -1 & 0 \\ -1/q & 1/q - 1 & 0 \\ 0 & 0 & (t-3)/(2-t) \end{pmatrix}.$$

For $H = U + V$, elimination in the first two rows of (3) yields

$$(4) \quad (4t^2 - 1)H'' + (8t^2 + 2t + 3)H' + (4t^2 + 4t + 7)H = 0,$$

whereas direct differentiation gives

$$(5) \quad (2-t)G' + (3-t)G = 0.$$

Thus both summands of the given generating function are D-finite.

We now combine them explicitly. Let $X = (U, V, G)^T$, let $r_0 = (1, 1, -1)$, and recursively set

$$r_{k+1} = r_k' + r_k A.$$

Then $F^{(k)} = r_k X$. Exact rational-function arithmetic gives the component-wise symbolic identity

$$(6) \quad p_3 r_3 + p_2 r_2 + p_1 r_1 + p_0 r_0 = (0, 0, 0),$$

where

$$\begin{aligned} p_3 &= (t+1)(2t-1)^2 = 4t^3 - 3t + 1, \\ p_2 &= (2t-1)(6t^2 + 4t + 1) = 12t^3 + 2t^2 - 2t - 1, \\ p_1 &= 3(4t^3 + 2t^2 + 2t - 5), \\ p_0 &= 4t^3 + 4t^2 + 3t - 18. \end{aligned}$$

It follows that

$$(7) \quad \mathcal{L}_3 F := p_3 F''' + p_2 F'' + p_1 F' + p_0 F = 0.$$

Equation (7) is already a polynomial-coefficient ODE proving D-finiteness. The next left multiple is chosen to expose the particular OEIS recurrence.

Proposition 2.1. *The generating function F satisfies*

$$\begin{aligned} 0 &= t(2t-1)F^{(5)} + (6t^2 + 10t - 2)F^{(4)} \\ &\quad + (6t^2 + 36t + 7)F^{(3)} + (2t^2 + 37t + 35)F'' \\ (8) \quad &\quad + (12t + 39)F' + 12F. \end{aligned}$$

Proof. With $D = d/dt$, define

$$\mathcal{M} = \frac{t}{(t+1)(2t-1)} D^2 + \frac{t-2}{(t+1)^2(2t-1)^2} D - \frac{1}{(t+1)^2(2t-1)^2}.$$

The Leibniz rule in right-normal form, $D^i f = \sum_{r=0}^i \binom{i}{r} f^{(r)} D^{i-r}$, gives the exact Ore-algebra identity

$$\begin{aligned} \mathcal{M} \circ \mathcal{L}_3 &= t(2t-1) \overset{5}{D} + (6t^2 + 10t - 2) \overset{4}{D} \\ &\quad + (6t^2 + 36t + 7) \overset{3}{D} + (2t^2 + 37t + 35) \overset{2}{D} \\ &\quad + (12t + 39) D + 12. \end{aligned}$$

The denominators in $\mathcal{M}$ are units at $t = 0$. Applying this identity to (7) proves (8). $\square$

3. COEFFICIENT EXTRACTION

For $m, j, k \geq 0$, the EGF convention gives

$$(9) \quad \left[\frac{t^m}{m!} \right] t^j F^{(k)} = (m)_j a_{m-j+k}, \quad (m)_j = m(m-1) \cdots (m-j+1).$$

Fix $n \geq 4$ and put $m = n - 4$. Applying (9) to (8) gives the complete coefficient collection below.

Term	Coefficient in m	Coefficient after $m = n - 4$
a_n	$-m - 2$	$2 - n$
a_{n-1}	$2m(m-1) + 10m + 7$	$2n^2 - 8n + 7$
a_{n-2}	$6m(m-1) + 36m + 35$	$6n^2 - 18n + 11$
a_{n-3}	$6m(m-1) + 37m + 39$	$(n-1)(6n-11)$
a_{n-4}	$2m(m-1) + 12m + 12$	$2(n-1)(n-2)$

Consequently, the coefficient of $t^{n-4}/(n-4)!$ in (8) is exactly the left-hand side of (2). This proves the recurrence for every $n \geq 4$. Finally, exact expansion at the origin gives

$$F(t) = \frac{t^2}{2!} + \frac{t^3}{3!} + 21 \frac{t^4}{4!} + 168 \frac{t^5}{5!} + 1968 \frac{t^6}{6!} + \cdots,$$

so the initial values and both indexing conventions in Theorem 1.1 follow. This completes the proof.

4. EXACT VERIFICATION AND LIMITATIONS

The accompanying certificate stores integer polynomial data for (6), the multiplier, the order-five operator, the recurrence, and initial coefficients. A standalone verifier using only exact rational arithmetic reconstructs the state system, the noncommutative operator product, the coefficient extraction, and the initial series. It imports no discovery code and rejects malformed or altered inputs. A separate direct symbolic differentiation check verifies both $\mathcal{L}_3 F = 0$ and $\mathcal{L}_5 F = 0$; exhaustive perfect-matching enumeration independently reproduces the combinatorial coefficients for $1 \leq n \leq 7$.

The theorem takes the published generating function (1) as its starting point; it does not re-prove the combinatorial derivation of that EGF. No

claim of minimal differential order is made, and the asymptotic formula on the OEIS page is outside the scope of this note. No proof assistant was used.

DATA, CODE, AND AI DISCLOSURE

The certificate, verifier, mutation tests, direct symbolic check, small exact enumerator, and a manifest of release hashes accompany this manuscript. The author used OpenAI Codex for literature searches, symbolic discovery, verifier and test implementation, manuscript drafting, and audit coordination. The decisive identities are exact and are independently reconstructed by the released standard-library verifier; no floating-point or probabilistic result is used as proof.

REFERENCES

1. Evgeniy Krasko and Alexander Omelchenko, *Enumeration of chord diagrams without loops and parallel chords*, The Electronic Journal of Combinatorics **24** (2017), no. 3, Paper No. P3.43, 23 pp.
2. OEIS Foundation Inc., *A278992: Number of simple chord-labeled chord diagrams with n chords*, The On-Line Encyclopedia of Integer Sequences, 2026, Accessed 2026-08-30.

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